CSE 220: Signals and Systems Sessional

Online – Fourier Series

Total Marks: 10 Time: 25 Minutes

Context & Objective

In Task 1 of the offline assignment, a 2D drawing is reconstructed using all $2N+1$ Fourier harmonics. However, for most closed curves, energy is concentrated within a subset of dominant harmonics.

Recall that the total energy of a periodic signal in terms of its Fourier coefficients is given by:

$$E_{\text{total}} = \sum_{n=-N}^{N} |c_n|^2$$

Your task is to implement energy-based harmonic pruning in `fs_redrawer.py` to compress the shape representation, and evaluate the resulting reconstruction fidelity.

Task 1: Energy-Preserving Harmonic Pruning

In the `FourierEpicycles` class, implement the method `prune_harmonics_by_energy`:

- Given a target energy ratio $r \in (0, 1]$, retain the minimal subset of most energetic harmonics such that their cumulative energy accounts for at least a fraction r of the total energy. Discarded harmonics should have their coefficients set to zero.
- The method should return the number of retained non-zero harmonics and the actual energy ratio achieved.

Task 2: Reconstruction Error Evaluation

In the `FourierEpicycles` class, implement the method `evaluate_reconstruction_error`:

- Compute and return the Mean Squared Error (MSE) between the ground-truth signal samples $f(t)$ and the reconstructed approximation $\hat{f}(t)$:

$$\text{MSE} = \frac{1}{M} \sum_{i=1}^{M} |f(t_i) - \hat{f}(t_i)|^2$$

Execution & Deliverables

In the main execution block of `fs_redrawer.py`, load `svgs/heart.svg` with $N = 150$. For target ratios $r \in \{0.96, 0.98, 0.99, 1.00\}$:

1. Perform harmonic pruning and compute the retained count, actual energy ratio, and reconstruction MSE.
2. Print the results on the console:

```
Target Ratio | Harmonics Retained | Actual Energy Ratio | MSE
------------------------------------------------------------------
 0.96        | ...                | ...                 | ...
 0.98        | ...                | ...                 | ...
 0.99        | ...                | ...                 | ...
 1.00        | 301                | 1.0000              | ...
```

3. Save the visual comparison plot as `heart_pruned_{r}.png`.